



School of Management
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Topic: Retail Waste & Profit Optimization

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Problem Statement

Retail businesses face significant losses due to unsold perishable inventory and inefficient pricing decisions. The challenge is to predict product-level waste and profitability, and identify key drivers to optimize inventory, pricing, and promotions.

Report Scope

- Explain the retail transaction dataset and the business meaning of its fields.
- Document the code workflow used for data inspection, EDA, business interpretation, and predictive modelling.
- Preserve the key EDA and model visualizations from the submitted PDF output.
- Translate the visual findings into operational decisions for inventory, pricing, promotions, and supplier/store management.

Dataset Explanation

The dataset contains transaction-level records for perishable retail products. Each row captures one product-store-supplier transaction with inventory quantity, demand, storage and handling conditions, expiry timing, pricing, promotion status, realized sales, waste, revenue, and profit.

Metric	Value
Rows	100,000
Columns	41
Transaction period	2023-01-01 to 2024-12-31
Unique products	63
Unique stores	50
Regions	5
Suppliers	20
Missing cells	0

Important Fields

Field Group	Columns	Business Role
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Product channel	and	Product_Id, Product_Name, Category, Store_Id, Region, Supplier_Id	Identifies where losses and profits occur.
Freshness and storage		Expiration_Date, Shelf_Life_Days, Days_Until_Expiry, Storage_Temp, Temp_Deviation	Measures spoilage exposure and cold-chain risk.
Demand handling	and	Daily_Demand, Demand_Variability, Distribution_Hours, Handling_Score, Packaging_Score	Explains sales movement and operational quality.
Pricing		Base_Price, Cost_Price, Discount_Pct, Selling_Price, Is_Promoted	Shows margin pressure and promotional behavior.
Outcomes		Units_Sold, Units_Wasted, Waste_Pct, Revenue, Waste_Cost, Profit, Profit_Margin_Pct	Defines the target performance to optimize.

Overall Business Snapshot

Measure	Value
Total units sold	20,283,565
Total units wasted	5,201,615
Average waste percentage	22.59%
Median waste percentage	12.40%
Total revenue	\$591.15M
Total waste cost	\$163.22M
Total profit	\$79.78M
Spoiled transaction rate	19.44%
Promoted transaction rate	14.90%

Code Workflow Summary

The notebook-style workflow started by loading the CSV dataset, checking structure and sample rows, validating missing values, and creating summary statistics. The analysis then moved into EDA, where waste, profit, expiry, category, region, discounting, sales, seasonality, and correlations were visualized. The final stage used predictive modelling outputs to identify the strongest drivers of waste.

Stage	What Was Done	Why It Matters
Data loading	Read transaction CSV and inspected shape, columns, head, and data types.	Confirms that the business fields needed for waste and profit analysis are present.
Data quality	Checked missing cells and field distributions.	Prevents inaccurate modelling caused by incomplete or malformed values.
EDA	Plotted waste distribution, expiry effect, discount effect, category and region waste, sales-waste, profit-waste, correlations, losses, and seasonality.	Turns raw operational data into interpretable business patterns.
Predictive modelling	Prepared feature inputs and reviewed feature importance for waste prediction.	Identifies the levers most useful for proactive intervention.
Business interpretation	Connected patterns to inventory sizing, markdown timing, promotion control, and product-level action.	Transforms analytics into retail decisions.

EDA and Business Interpretation

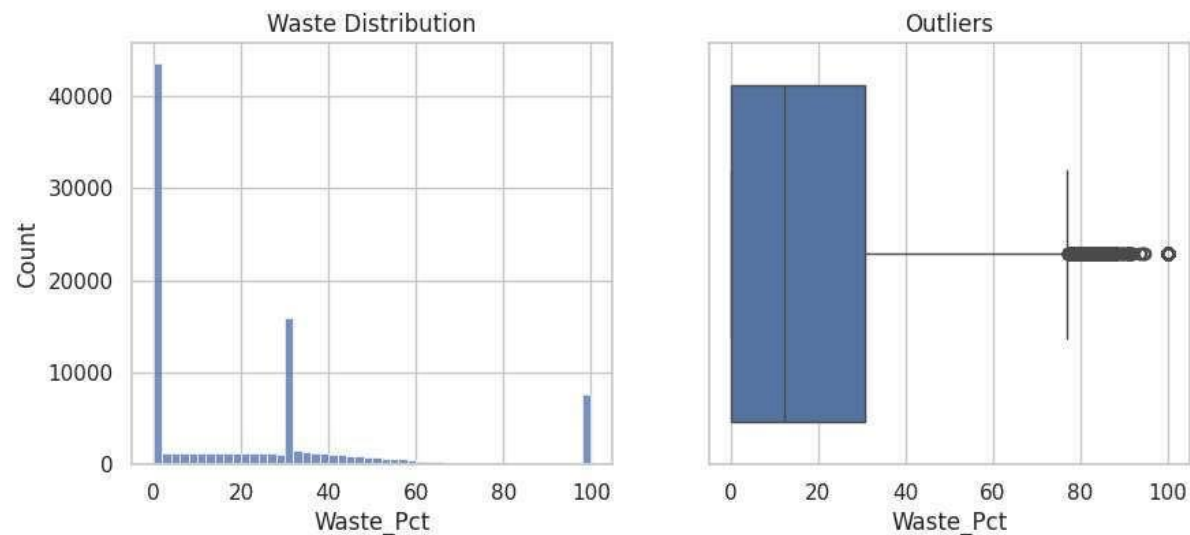


Figure 1. Waste distribution and outlier behavior.

Waste is heavily concentrated at low percentages for many transactions, but there are visible spikes and extreme outliers near very high waste levels. Business interpretation: most inventory is controllable, but a smaller set of product-store situations can destroy margin quickly and should be flagged for exception handling.

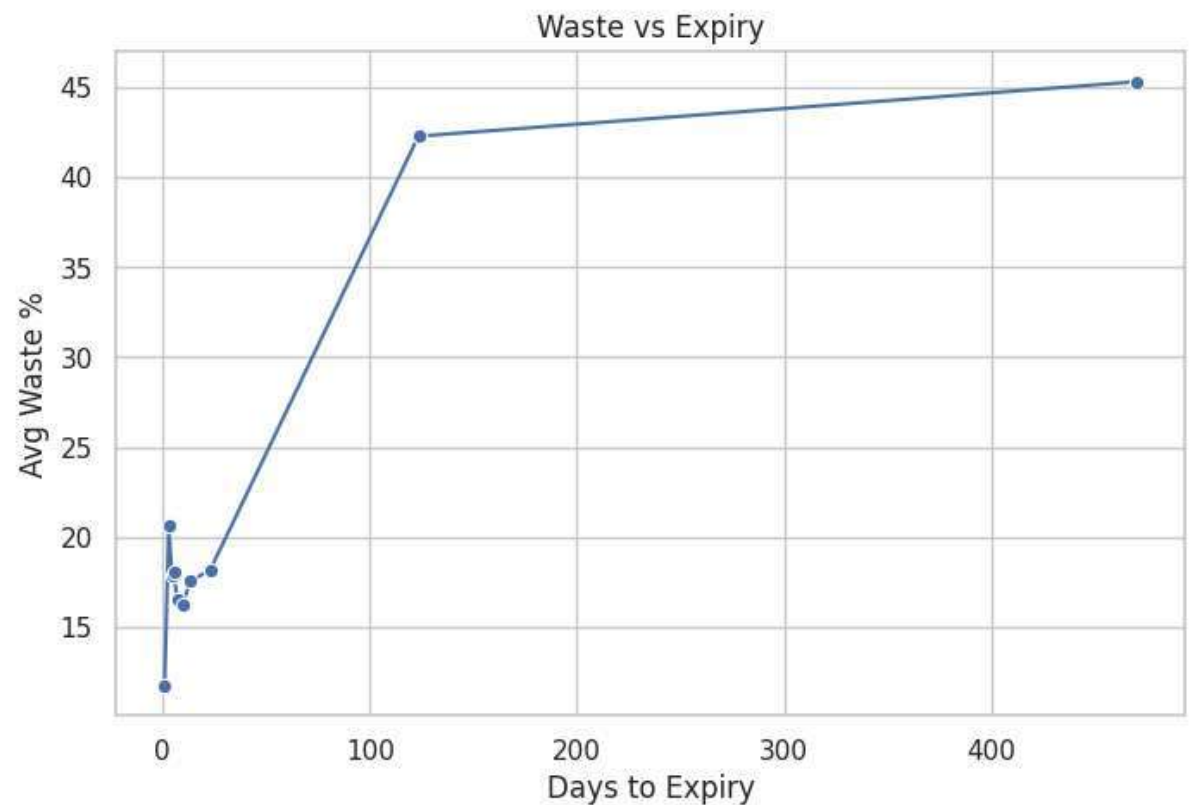


Figure 2. Average waste percentage by days to expiry.

The expiry relationship shows that waste changes materially across freshness windows. Longer shelf-life and expiry timing are associated with different waste levels, indicating that replenishment rules should be product-specific rather than uniform across all categories.

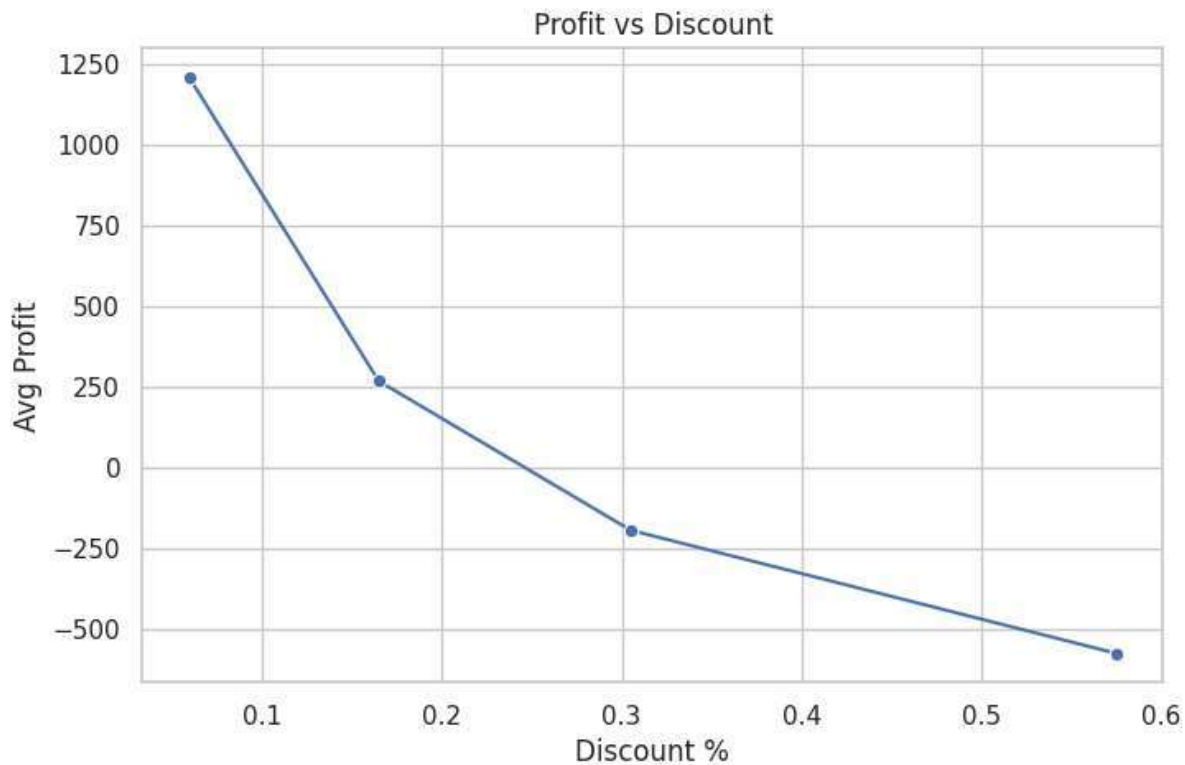


Figure 3. Average profit by discount percentage.

Profit falls sharply as discount percentage increases. Small discounts preserve positive average profit, while aggressive markdowns move the average into negative territory. Business interpretation: discounts should be targeted by product risk and timing, not applied as broad blanket markdowns.

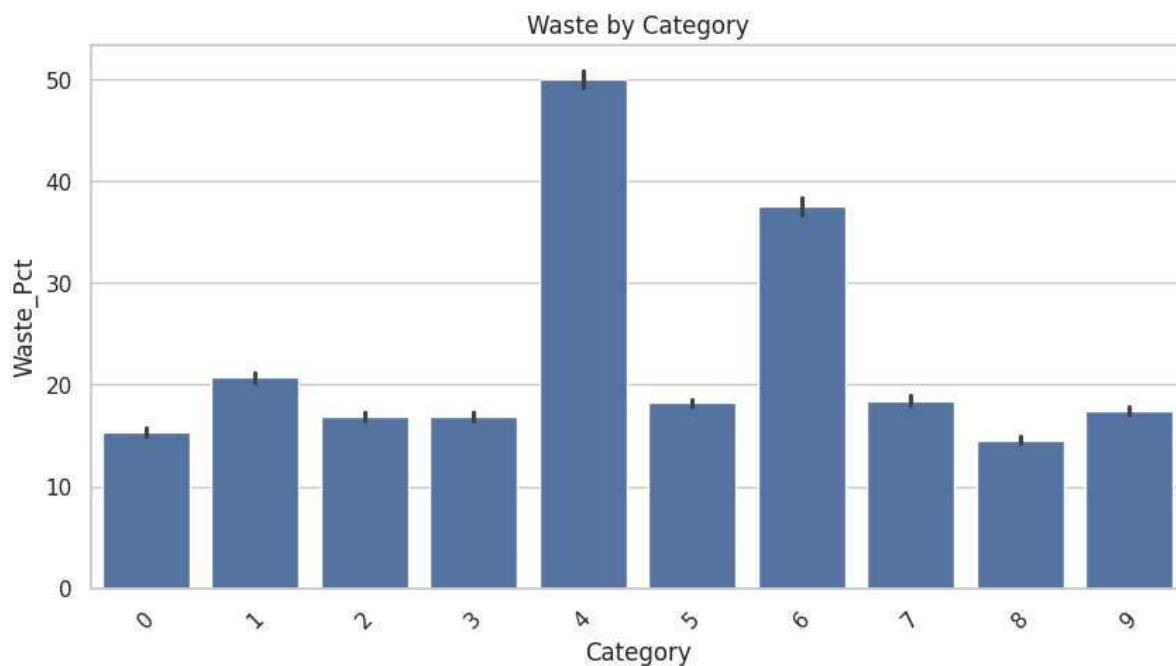


Figure 4. Waste percentage by category.

Category-level waste differs strongly. Frozen Meals and Pharmaceuticals stand out in the dataset summary, with average waste of 50.01% and 37.49% respectively. Business interpretation: category-specific policies are necessary because the same stocking and promotion rule will not fit every product family.

Category Performance

Category	Avg Waste %	Units Wasted	Profit
Frozen_Meals	50.01%	1,167,759	-\$789.5K
Pharmaceuticals	37.49%	947,716	\$39.52M
Beverages	20.70%	418,323	\$3.87M
Produce	18.45%	400,822	\$2.80M
Meat	18.20%	423,986	\$7.82M
Seafood	17.41%	410,919	\$9.43M
Dairy	16.91%	332,290	\$5.04M
Deli	16.90%	375,593	\$6.94M
Bakery	15.35%	365,116	\$3.31M
Ready_to_Eat	14.56%	359,091	\$1.84M

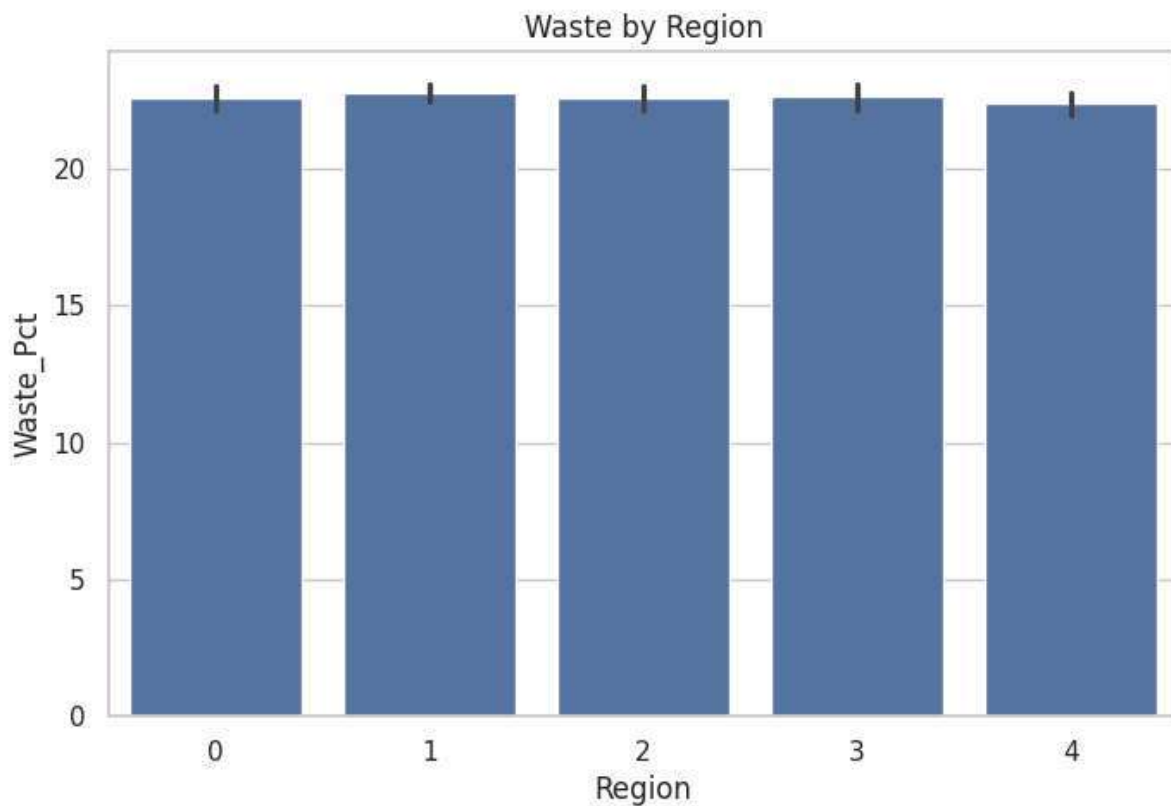


Figure 5. Waste percentage by region.

Regional waste is relatively flat, with all regions near the overall average. Business interpretation: the biggest opportunities appear to be product/category and inventory-policy driven rather than purely geographic.

Regional Waste Summary

Region	Avg Waste %	Units Wasted	Profit
Northeast	22.73%	1,564,404	\$23.72M
Southwest	22.63%	728,894	\$10.58M
Southeast	22.58%	1,044,528	\$16.51M
Midwest	22.58%	830,099	\$13.49M
West	22.37%	1,033,690	\$15.49M

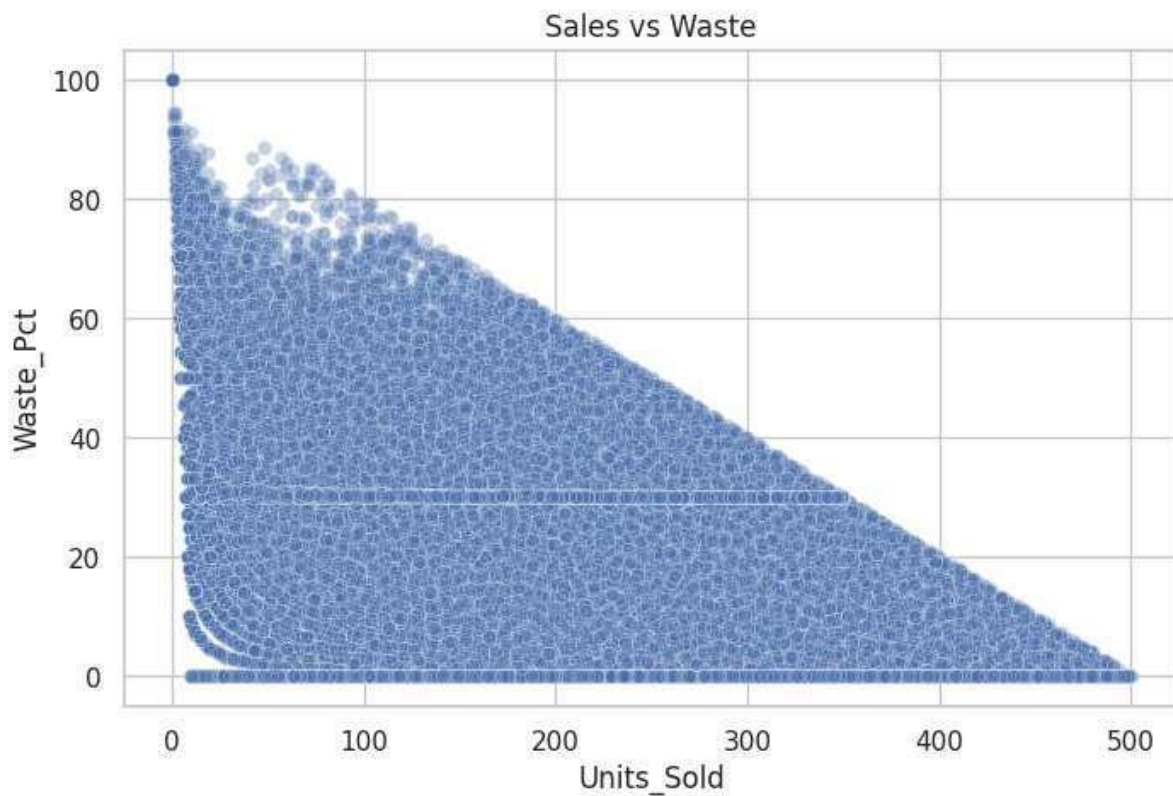


Figure 6. Sales volume versus waste percentage.

The scatter pattern shows a strong negative relationship: higher units sold generally correspond to lower waste percentage. Business interpretation: low-velocity items are the main candidates for smaller order quantities, dynamic markdowns, or demand stimulation.



Figure 7. Profit versus waste percentage.

Profitability deteriorates as waste rises. High-waste records include substantial negative profit cases, confirming that waste is not only an operational issue but a direct margin issue.

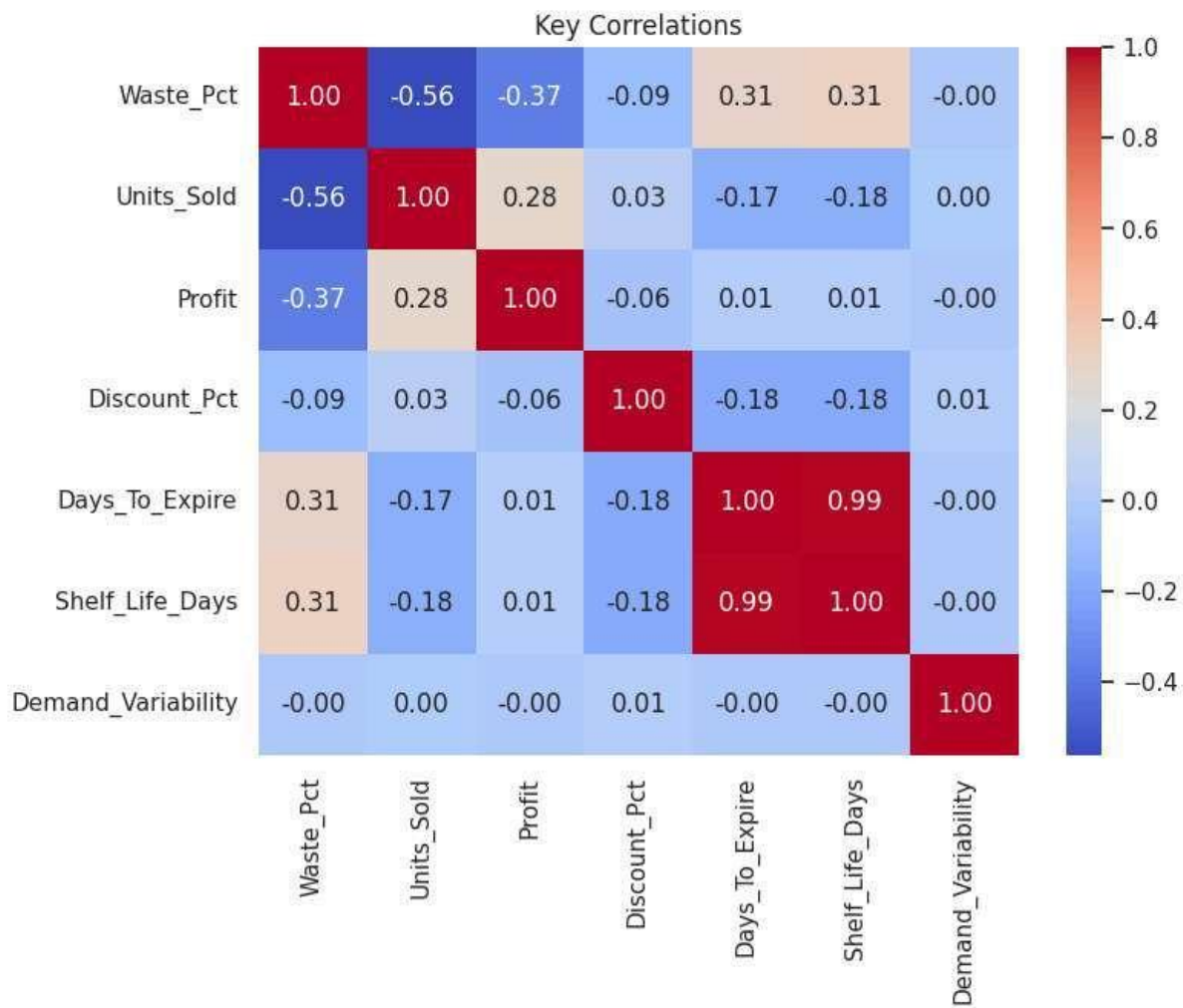


Figure 8. Correlation matrix for key variables.

The heatmap confirms the most important directional relationships: waste is negatively correlated with units sold (-0.56) and profit (-0.37), while days to expiry and shelf life are strongly related to each other. Business interpretation: sales velocity and freshness timing should be primary signals in a waste-risk alert system.

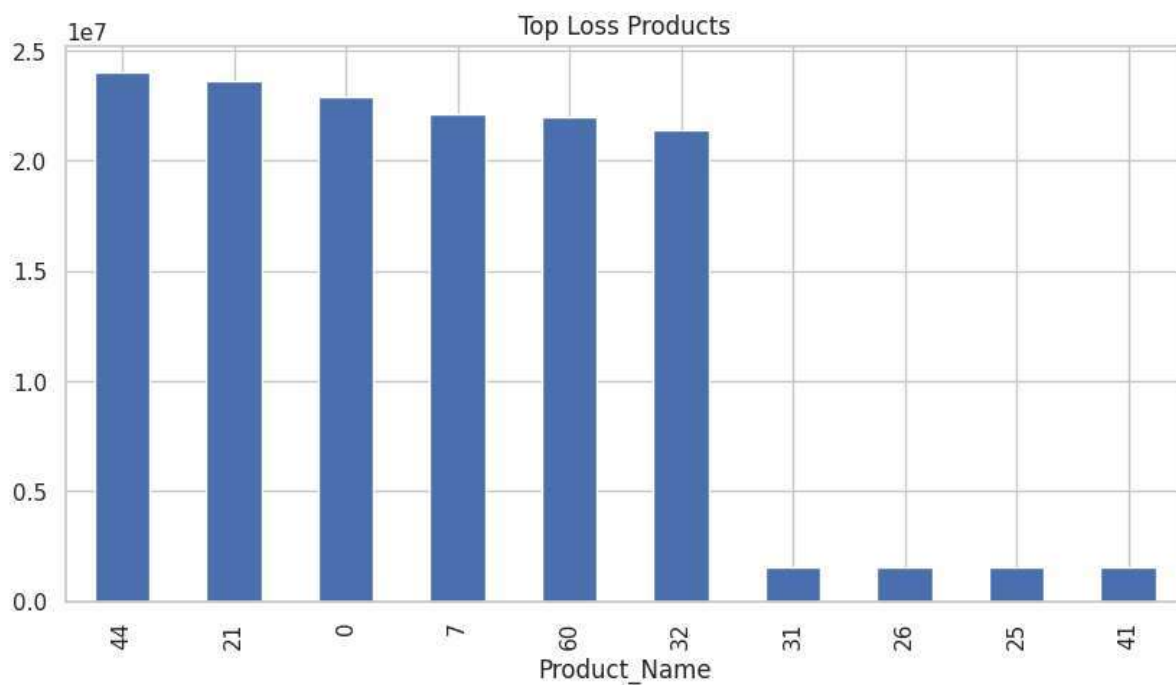


Figure 9. Top loss products.

The loss-product view isolates where financial leakage is concentrated. In the dataset summary, Frozen Vegetables, Frozen Desserts, Pizza, Frozen Dinners, and Ice Cream are the most negative-profit product names.

Top Loss Products

Product	Total Profit	Avg Waste %	Units Wasted
Frozen Vegetables	-\$257.7K	50.55%	231,069
Frozen Desserts	-\$171.5K	50.48%	235,570
Pizza	-\$145.7K	48.98%	231,544
Frozen Dinners	-\$115.9K	50.91%	235,862
Ice Cream	-\$98.8K	49.15%	233,714
Sandwiches	\$265.5K	14.84%	61,782
Salads	\$288.6K	14.26%	56,574
Fresh Pasta	\$294.3K	14.04%	56,839
Wraps	\$301.6K	14.87%	61,631
Sushi	\$340.5K	14.55%	58,297

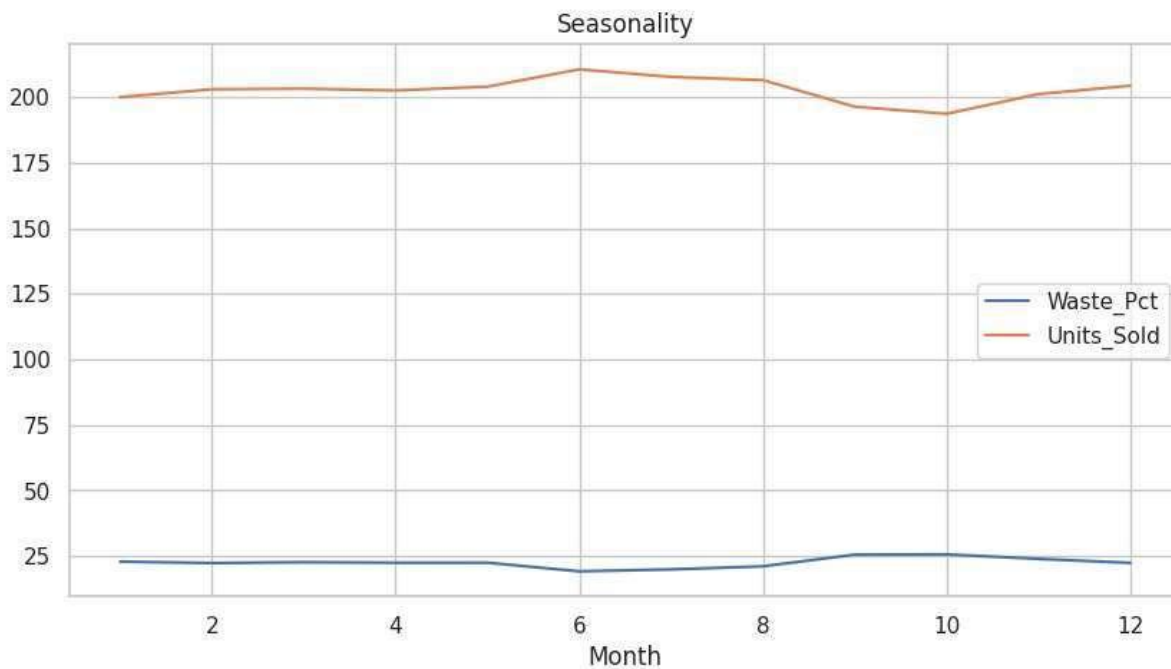


Figure 10. Seasonality in waste and units sold.

Seasonality appears moderate, with units sold and waste changing across months. Business interpretation: replenishment and promotion calendars should account for monthly demand shifts, especially before periods where sales soften and waste rises.

Predictive Analytics Findings

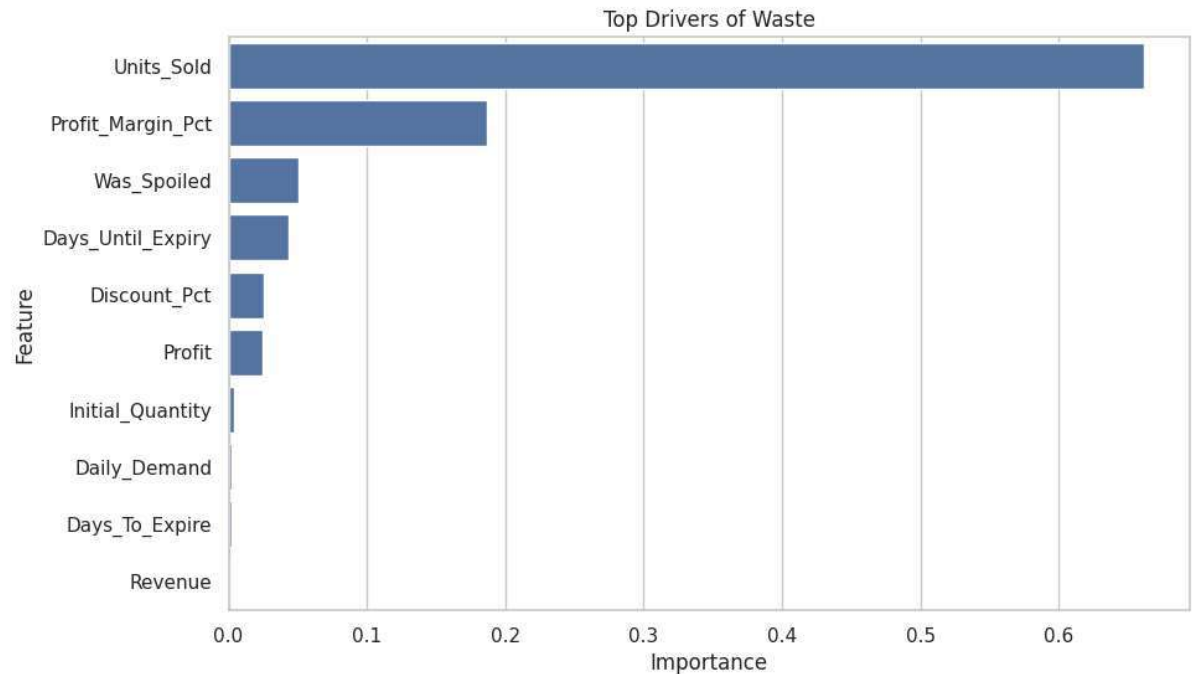


Figure 11. Top model drivers of waste.

The feature-importance output shows Units_Sold as the dominant driver, followed by Profit_Margin_Pct, Was_Spoiled, Days_Until_Expiry, Discount_Pct, and Profit. Business interpretation: the most valuable prediction system should combine sales velocity, margin protection, spoilage status, expiry timing, and discount strategy.

Model Use in Business

- Predict high-waste product-store combinations before the expiry window closes.
- Prioritize interventions for items with low sales velocity and shrinking margin.
- Recommend smaller replenishment quantities for recurring high-waste product groups.
- Trigger markdowns only when expected waste reduction outweighs profit loss from discounting.
- Surface product-level loss leaders for assortment review or supplier negotiation.

Recommendations

Area	Recommended Action	Expected Impact	Business
Inventory planning	Reduce replenishment quantity for slow-moving, high-waste SKUs and use category-specific safety stock.	Lower waste cost without starving high-velocity products.	
Markdown strategy	Use controlled, time-based discounts instead of broad high markdowns.	Protects profit while still clearing at-risk inventory.	
Category management	Prioritize Frozen Meals and other high-waste categories for assortment and ordering review.	Targets the largest waste percentages first.	
Store operations	Monitor spoilage, handling score, packaging score, and temperature deviation as exception signals.	Reduces avoidable waste from process failures.	
Predictive alerts	Deploy a waste-risk score using sales velocity, margin, expiry timing, spoilage flag, discount, and profit signals.	Moves the business from reactive write-offs to proactive prevention.	

Conclusion

The analysis shows that waste and profit are tightly connected. High waste reduces profitability, while sales velocity, margin, spoilage status, expiry timing, and discounting explain much of the operational risk. The strongest business opportunity is to replace uniform ordering and discounting rules with product-level predictive decisions that balance waste avoidance against profit protection.